

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

One potential explanation for the website's connection timeout error message is a DoS attack.

There was a large volume of incoming HTTPS TCP connections originating from the IP address 203.0.113.0, involving SYN packets sent to the server and the server's corresponding responses. Ultimately, the server crashed because the volume of requests exceeded the port's capacity. This incident may have been a SYN flood attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The client sends a SYN packet to the web server to request to connect.
2. The destination replies to the source with a SYN-ACK packet to accept the connection request. The destination will reserve resources for the source to connect..
3. A final ACK packet is sent from the source to the destination acknowledging the permission to connect.

When a malicious attacker sends a large number of SYN packets simultaneously, the server responds to these requests.

This is confirmed by the activity shown in the logs. The logs reveal a large volume of incoming HTTPS-related TCP connection attempts—specifically SYN packets sent to the server from the IP address 203.0.113.0—along with the server's responses. The volume of requests exceeds the port's capacity, ultimately causing the server to crash.

In a SYN flood attack, the attacker rapidly sends a large volume of SYN packets, exhausting the server's available resources reserved for connections. Once this happens, the server is unable to allocate resources for legitimate TCP connection requests.

Logs indicate that the web server is overwhelmed and unable to process visitors' SYN requests. The server cannot establish connections for new visitors, resulting in connection timeout errors for them.